Mohamed Helmy

Software Engineer | Flutter Developer

moohaamed.hhelmy.13@gmail.com +20 1140080928 Benha, Egypt

linkedin.com/in/mohamedHelmy github.com/MooHelmy

PROFILE

Aspiring Mobile Developer with strong expertise in the Android framework and Flutter SDK. Skilled in building scalable applications using Clean Architecture, MVVM, and Bloc State Management. Proven track record of integrating advanced features such as Machine Learning (CNNs) and IoT systems into mobile platforms. Eager to leverage skills in OOP, SOLID principles, and problem-solving to contribute to innovative development teams. Seeking an opportunity to contribute to innovative projects, grow as a developer, and learn from industry experts.

EDUCATION

Bachelor of computer science 2020 – 2024

Benha University - Faculty of Computers and Artificial Intelligence Benha, Egypt

Computer Science major,

GPA: 3.6,

Graduation Project: Excellent

PROFESSIONAL EXPERIENCE

Software Engineer 12/2023 – Present

Cairo, Egypt

- **Flutter Developer (Freelancer):**
  Designed and built multiple cross-platform mobile applications using **Flutter and Dart**,, each customized to meet the unique UI/UX, feature, and performance requirements of various clients, ensuring a scalable and performant codebase.

12/2024 – 12/2025

Cairo, Egypt

- **Software Engineer (Military Service):**
  MEMS Project: Engineered a high-security Flutter application to digitize military educational logistics and lecture tracking. Process Automation: Replaced manual workflows with an automated Attendance & Analytics module, increasing reporting accuracy and operational speed. Data Integrity: Developed a robust Offline-First synchronization layer tailored for sensitive, restricted-connectivity environments.

7/2023 – 10/2023

Benha, Egypt

- **Mobile Application Development Trainee:**
  Information Technology Center for Research and Consulting (*ITCRC*)
  Building cross-platform applications using Flutter and Dart. Implementing MVVM and Clean Architecture for scalable mobile apps. Integrating RESTful APIs and Firebase services. Applying state management techniques (Cubit, Provider). Receiving training in Business English, Leadership, and Soft Skills to enhance professional communication and teamwork.

01/2023 – 04/2023

- **Flutter Trainee Information Technology Institute– ITI** Completed intensive Flutter training with hands-on projects. Gained experience in app development, API handling, and Git version control..

SKILLS

Programming Languages
Dart, Java, C++, C, SQL, HTML, Assembly
Python, Kotlin

Frameworks
Flutter, MVVM, Clean Architecture, Android APIs, Android SDK, Google Maps API, Dio, http Provider, Block/Cubit, Firebase Realtime DB / Firestore

Database
Hive, Server, My SQL Oracle, Firebase

Technologies
Sessions, Cookies, Firestore Authentication, Version Control (Git & GitHub), AI Model Deployment, NLP, Computer Vision, Embedded Systems, Machine Learning, IoT

Additional Skills
OOP, SOLID Principles, Design Patterns, Data Structure, Algorithms, ML, Deep Learning (CNN), Unit Testing, Git, AI Models, Computer Vision, Embedded Systems, IOT

PROJECTS

Scan Care (Graduation Project) | MVVM

Tools: Flutter, Python, CNN, Firebase, Clean Architecture, Bloc State Management, Google Maps Services, PayMent(Stripe)

02/2024– 06/2024

- Developed a medical diagnostic mobile application utilizing Convolutional Neural Networks (CNNs) to analyze medical images and detect health issues.
- Implemented secure patient-provider communication, streamlined appointment booking, and integrated reliable online payment gateways.
- Integrated Google Maps services to direct users to nearby clinics and healthcare facilities.
- Built using Clean Architecture with Bloc State Management to ensure code scalability and maintainability.
- Integrated 5 AI models, automating 60% of tasks.

Bookly App | API , MVVM

12/2023 –02/2024

Tools: Dart, MVVM, Bloc, APIs

- *Engineered a book discovery application allowing users to search and read from a vast collection of books.*
- *Implemented complex state management using Bloc and utilized RESTful APIs for efficiently fetching and displaying book data.*
- *Built using the MVVM (Model-View-View Model) architectural pattern to ensure separation of concerns and testability.*

Online Shopping App | API , MVVM

12/2023 – 01/2024

Tools: Dart, Firebase, MVVM ,Api ,PayMent(Stripe)

- *Created a comprehensive e-commerce platform for fashion and electronics featuring cart management, a favorites system, and detailed transaction history.*
- *Structured the application using the MVVM (Model-View-View Model) architectural pattern for modularity.*
- *Leveraged Firebase for real-time data synchronization, authentication, and backend services.*

Mobile Store Management Application | MVVM 09/2023 - 10/2023

Tools: Dart, MVVM, Bloc State Management, Firebase

- *Developed a comprehensive mobile store management application providing distinct feature sets for administrators and customers.*
- *For Administrators: Implemented full CRUD (Create, Read, Update, Delete) functionality for managing product inventory, viewing detailed user records, and managing invoice processing.*
- *For Customers: Enabled users to browse the latest product offerings, manage purchases, and utilize a dedicated maintenance request section.*
- *Structured the application using the MVVM (Model-View-ViewModel) architectural pattern and Bloc State Management to ensure scalability and predictable state control.*
- *Utilized Firebase for robust authentication, real-time data synchronization, and reliable backend service delivery*

05/2024 - 06/2024

IoT Smart Home System

Tools: Arduino, Python, Firebase, DHT11 (Sensor Integration)

- *Designed and built an IoT-based smart home dashboard to remotely control various household appliances (e.g., lamps, fans) and display real-time environmental data.*
- *Integrated sensors, including DHT11 (for temperature/humidity), fire detection, and gas leak monitoring, to enhance home security and energy efficiency.*
- *Implemented voice control features for seamless, hands-free operation of lighting systems*

INVOLVEMENT & INTERNSHIPS

Android developer
ITCRC 7/2022- 10/2022
Benha, Egypt

Mobile development using (Flutter)
ITCRC 7/2023 - 10/2023
Benha, Egypt

Flutter developer
Information Technology Institute– ITI 01/2023 - 04/2023
Menoufia, Egypt

YOUTUBE CHANNEL

🎥 @eng_mo723

Created and shared 100+ educational videos on software engineering, Assembly, and embedded systems.
Built an engaged community with over 50,000 total views.
Example playlists:
- Assembly Development
- Embedded Systems
- Career in Google Cloud

SOFT SKILLS

Communication | Teamwork | Adaptability | Creativity | Time Management | Leadership